

Weather Insights API — Reference

API version 1.3

Overview

The Weather Insights API provides current conditions, hourly forecasts, and historical weather data for any latitude/longitude coordinate worldwide. All endpoints return JSON and require an API key passed via the X-API-Key header.

Authentication

Every request must include a valid API key. Keys are generated from the developer dashboard and are tied to a specific rate-limit tier. Requests without a valid key return HTTP 401.

Rate Limits

Tier	Requests/min	Requests/day
Free	10	1,000
Growth	60	50,000
Enterprise	600	Unlimited

Requests exceeding the per-minute limit receive HTTP 429 with a Retry-After header indicating seconds until the limit resets.

Endpoint: GET /v1/current

Returns current weather conditions for a coordinate.

- **lat** (float, required) — latitude, -90 to 90
- **lon** (float, required) — longitude, -180 to 180
- **units** (string, optional) — "metric" (default) or "imperial"

Response fields: temperature, feels_like, humidity_pct, wind_speed, condition_code, observed_at (ISO 8601).

Endpoint: GET /v1/forecast/hourly

Returns an hourly forecast for the next 48 hours.

- **lat, lon** (float, required) — as above
- **hours** (int, optional) — number of hours to return, 1–48, default 24

Endpoint: GET /v1/historical

Returns daily aggregated weather for a past date range, up to 5 years back.

- **lat, lon** (float, required) — as above
- **start_date, end_date** (string, required) — ISO 8601 dates, max 90-day range per request

Error Codes

- **400** — malformed parameters (e.g. latitude out of range)

- **401** — missing or invalid API key
- **404** — no data available for the given coordinate
- **429** — rate limit exceeded
- **500** — internal server error; safe to retry with exponential backoff